

Prequential Streaming Evaluation of Stateful Time-Series Forecasters with Online Conformal Calibration: A Diagnostic Atlas and the `preqts` Harness

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github.com/fsantibanezleal/CAOS_RES_ChronoScope

Abstract. Real forecasting is streaming: observations arrive one by one, the model updates, and its prediction intervals must stay calibrated as the series drifts. Yet every widely used forecasting benchmark harness evaluates models statelessly on rolling windows, carrying no state across windows and never running a test-then-train loop. We address this gap with a prequential streaming evaluation harness for *stateful* forecasters, released as the open-source package `preqts`: it implements Dawid’s predict-then-observe-then-update protocol behind a `StatefulForecaster` interface with an explicit covariate-arrival policy, and records four trajectories per method, rolling scaled error, rolling empirical coverage, the rolling weighted quantile loss (a quantile-based probabilistic score), and cumulative compute cost, the last of which separates a stateful constant-cost model from a replay-everything wrapper. The harness is embedded in an atlas that first *understands* each series with ten diagnostic families (stationarity, autocorrelation, seasonality, filters, change points, volatility, distribution and complexity, fractal structure, nonlinear dynamics, and cross-series causality) and then forecasts it with a nineteen-method ladder (classical, statistical, gradient-boosted, deep, and stateful foundation models), so that the diagnosis explains the no-free-lunch leaderboard. Using the harness we test a concrete claim: that online conformal calibration keeps streaming intervals at nominal under distribution shift. On fifteen series that span distinct regimes and include white-noise and random-walk negative controls, wrapping a Theta point forecaster in Adaptive Conformal Inference or Conformal-PID control reduces the mean absolute coverage error at the nominal 80% level from 0.042 (raw Theta) and 0.060 (seasonal-naive) to 0.0057 and 0.0060, a seven-to-tenfold reduction, and shrinks the across-case span of the final coverage from 0.19–0.40 to 0.04, without changing the point forecast. The calibration holds on the negative controls, where a spurious skill gap would flag harness leakage. The contribution is the reusable stateful-streaming harness, evaluated under the protocol forecasters are actually deployed under, and a leakage-guarded, reproducible demonstration of where online conformal calibration earns its keep.

Index Terms. time-series forecasting, prequential evaluation, streaming, conformal prediction, adaptive conformal inference, calibration, coverage, stateful forecasters, foundation models, diagnostics, benchmark, reproducibility.

I. INTRODUCTION

FORECASTING in production is a streaming problem: a datum arrives, the model issues a prediction for the next horizon, the truth is later revealed, and the model updates its state before the next step; the prediction interval must remain calibrated throughout, even as the series drifts across regimes. This is not how forecasting models are usually evaluated. The widely used harnesses, including the rolling-window backtests of GluonTS and Darts and the window-based fev-bench [4], score forecasters *statelessly*: each window is independent, no learned state carries across windows, and there is no test-then-train loop. None of these harnesses evaluates a *stateful* streaming forecaster prequentially with an explicit policy for when covariates arrive.

We fill it with `preqts`, an open-source harness implementing the prequential principle, Dawid’s “predictive sequential” test-then-train protocol [1]: at each step the model predicts, the truth is revealed and scored, and only then does the model update. Skill becomes a trajectory over the stream rather than a single number. The harness is embedded in an atlas (Section II) that pairs the forecast with a diagnostic layer, so that the character of a series explains why one method family wins and another fails, the no-free-lunch principle, made legible. We then use the harness to answer a specific question (Section V): does online conformal calibration keep streaming intervals calibrated under distribution shift? Our contributions are: (i) the reusable prequential streaming harness `preqts` for stateful forecasters with a covariate-arrival policy; (ii) a reproducible, control-guarded demonstration that online conformal calibration reduces streaming coverage error seven-to-tenfold and collapses its across-case spread, without changing the point forecast.

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The harness is released as the open-source package `preqts` (<https://pypi.org/project/preqts/>); the atlas (code, per-case artifacts, workbench) is at https://github.com/fsantibanezleal/CAOS_RES_ChronoScope (MIT), and an interactive workbench is available at <https://chronoscope.fasl-work.com>.

II. THE ATLAS: UNDERSTAND, THEN FORECAST

ChronoScope pairs a diagnostic layer with a forecast ladder over fifteen cases, each computed deterministically (an artifact is a pure function of case and seed). **Ten diagnostic families** characterize a series before it is forecast: stationarity and unit roots (ADF, KPSS, Phillips–Perron, DF-GLS, Zivot–Andrews); autocorrelation (ACF/PACF, Ljung–Box); seasonality (periodogram, STL/MSTL, seasonal strength); filters and adaptive decompositions (Hodrick–Prescott, Baxter–King, EMD/CEEMDAN, wavelet scalogram); change points and regimes (PELT, OLS-CUSUM, Hamilton Markov-switching); volatility and transforms (ARCH-LM, GARCH, Box–Cox); distribution, normality and complexity (moments, Jarque–Bera, sample/spectral entropy, the BDS test, and the catch22 feature set [9]); fractal and multifractal structure (Hurst via R/S and DFA, MF-DFA, ARFIMA long memory); nonlinear dynamics and chaos (Takens embedding, largest Lyapunov exponent, recurrence quantification, the 0–1 test, with a surrogate-data test for nonlinearity); and cross-series causality (cross-correlation lead/lag, Granger, Engle–Granger and Johansen cointegration). Each is a verified implementation wrapped in a NaN-safe API and precomputed per case.

A **nineteen-method forecast ladder** is then scored on the same cases: classical numpy live methods (seasonal-naive, SES, Holt, Holt–Winters, Theta [8]), statistical methods (AutoARIMA, AutoETS, AutoTheta), a gradient-boosted method (LightGBM), deep methods, and stateful zero-shot foundation models (Chronos [10], TimesFM [11]). The cases span distinct regimes by construction, level shift, deterministic chaos, GARCH heteroskedasticity, intermittent demand, long memory, coupled daily/weekly seasonality, exogenous promotions, plus real CC-BY series (UCI Electricity, Beijing PM2.5, M4 excerpts [6]) and two negative controls, white noise and a near-random walk. There is no universal winner: the diagnosis explains the leaderboard, which is exactly the setting in which leakage-free scoring matters most.

III. THE PREQUENTIAL STREAMING HARNESS

Let a stream deliver observations $y_1, y_2, \dots$. At step t a stateful forecaster f with state s_t issues a horizon- h prediction and an interval at nominal level $1 - \alpha$; the truth y_{t+h} is then revealed, the prediction scored, and the state updated (Fig. 1),

$$(\hat{y}_{t+h}, \hat{I}_t) = f_{\text{predict}}(s_t), \quad s_{t+1} = f_{\text{update}}(s_t, y_{t+h}). \quad (1)$$

The update strictly follows scoring, so no point is used to predict itself; a covariate-arrival policy declares, per exogenous channel, whether its value at $t+h$ is known at prediction time (a planned promotion) or revealed with the target (a realized driver). After a warmup the harness records, as functions of stream position, four trajectories: the rolling scaled error, the rolling empirical coverage, the rolling weighted quantile loss, and the cumulative compute cost, the constant-cost-per-step signature that separates a stateful model from a replay-everything wrapper. Point accuracy uses the mean absolute scaled error (MASE) [5]; interval quality is read two ways that cannot be gamed independently, empirical coverage versus the nominal level, and the weighted quantile loss (WQL),

the quantile (pinball) loss summed over the levels 0.1, 0.5 and 0.9 and normalized by the sum of absolute target values, which penalizes both interval width and misses. The quantile loss is a proper scoring rule for quantile forecasts [7], and its average over quantile levels approximates the continuous ranked probability score (CRPS) [10]. The harness sits behind a `StatefulForecaster` interface so classical methods, gradient-boosted models, and stateful foundation models are evaluated identically, and it is published as the package `preqts`; because the streaming artifact carries only aggregate metric trajectories (no raw series values are recoverable), the demonstration can be released even for real datasets licensed for local use only.

IV. ONLINE CONFORMAL CALIBRATION

Split-conformal prediction issues an interval whose width is a quantile of past nonconformity scores (here absolute residuals); under exchangeability its coverage is $1 - \alpha$. Streams are not exchangeable, so the effective level must adapt. **Adaptive Conformal Inference** [2] updates the level each step by the observed miss,

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = \mathbf{1}\{y_{t+h} \notin \hat{I}_t\}, \quad (2)$$

so long-run coverage converges to the target under arbitrary distribution shift. **Conformal-PID control** [3] generalizes this to a proportional-integral-derivative controller on the coverage error, with ACI as the integral-only special case; the proportional and derivative terms buy faster coverage recovery after a regime change. Both wrap any point forecaster and change only the interval, never the point prediction.

V. RESULTS

The headline is Fig. 2. Averaged over the fifteen cases, the raw Theta intervals miss the nominal 80% target by a mean absolute coverage error of 0.042 and the seasonal-naive floor by 0.060, with worst-case per-case errors of 0.11 and 0.31. Wrapping the *same* Theta point forecaster in ACI or Conformal-PID cuts the mean coverage error to 0.0057 and 0.0060, a sevenfold reduction over raw Theta and tenfold over seasonal-naive. The across-case view (Fig. 3) shows the same effect as a collapse of dispersion: the final coverage of the calibrated methods clusters within a 0.04 span at nominal, while raw Theta spreads over 0.19 and seasonal-naive over 0.40. The point forecast is unchanged (the MASE of Theta, Theta+ACI and Theta+PID is identical per case); only the interval self-corrects. The effect is regime-robust: where raw intervals drift most, the calibrators pull the time-averaged coverage back (intermittent demand from 0.91 raw to 0.81 with ACI; chaotic Mackey–Glass from 0.77 raw to 0.80; real UCI electricity from 0.76 raw to 0.80); and on the negative controls calibration lifts the white-noise coverage from 0.75 (raw) to 0.79 while the near-random-walk control is already at ≈ 0.80 raw and calibration leaves it there (0.78), so the harness does not manufacture skill on unpredictable series, exactly the leakage guard the controls exist to enforce.

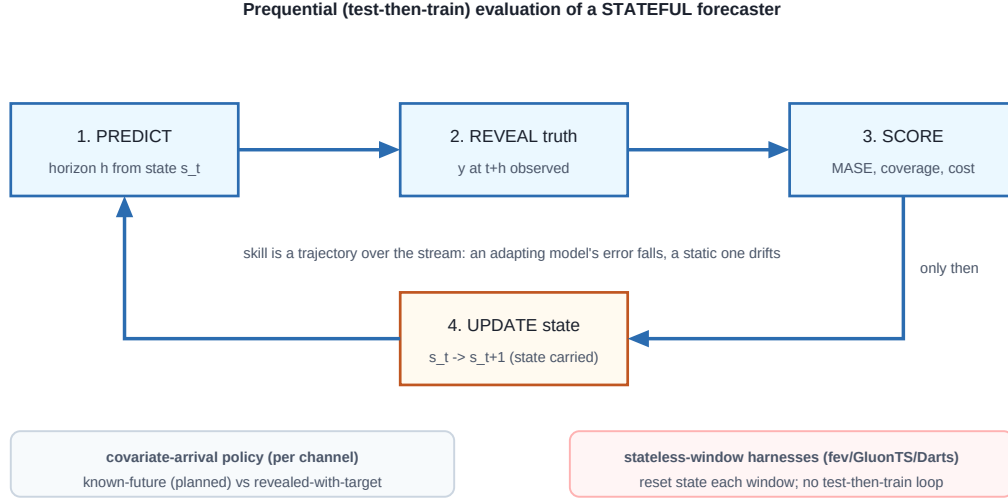


Fig. 1. The prequential test-then-train loop for a stateful forecaster. At each step the model predicts from its current state, the truth is revealed and scored, and only then is the state updated; a covariate-arrival policy declares, per channel, whether a value is known at prediction time or revealed with the target. Stateless-window harnesses reset state each window and run no test-then-train loop, so they cannot see a calibrator recover coverage after a shift.

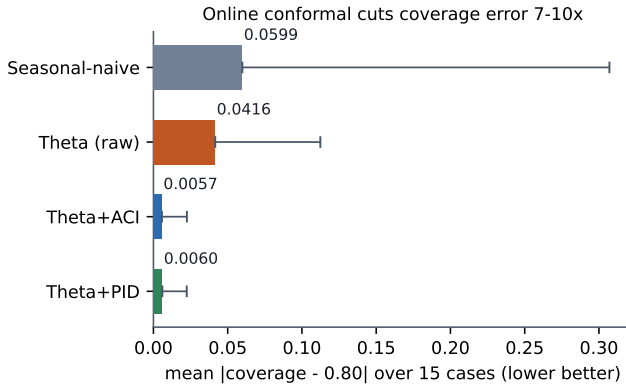


Fig. 2. Mean absolute coverage error at nominal 80% over the fifteen cases (bar) with the worst-case per-case error (whisker). Online conformal cuts the error seven-to-tenfold. Data: the per-case `streaming.json` artifacts.

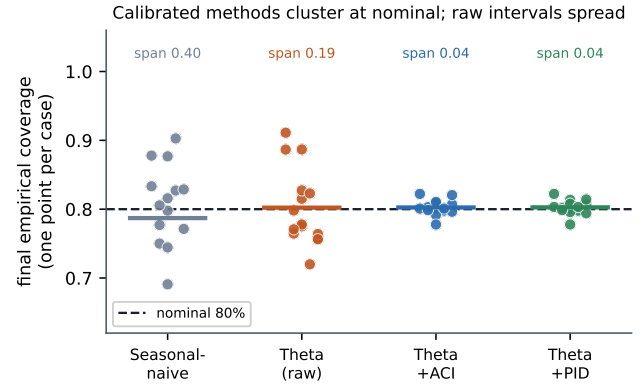


Fig. 3. Final empirical coverage, one point per case. The calibrated methods cluster tightly at the nominal level (across-case span 0.04) while the raw and naive intervals spread widely (span 0.19 and 0.40). Data: `streaming.json`.

VI. DISCUSSION AND SCOPE

That online conformal calibration keeps streaming intervals near nominal under shift is not by itself surprising, it is what ACI and Conformal-PID were designed to do. The contribution is that it is measured under the correct protocol. A stateless rolling-window harness cannot see a calibrator recover coverage after a regime change, because it does not carry the calibrator's state across windows; the prequential harness does, and it charges the compute cost that distinguishes a constant-cost stateful method from a replay wrapper. The negative controls and the identical point forecasts across raw and calibrated variants rule out the two easy ways such a result could be an artifact, harness leakage and a point-accuracy confound. Scope: the demonstration uses a Theta base forecaster because it is a strong classical method that runs live; the calibration is method-agnostic and wraps any forecaster. The atlas mixes seeded synthetic regimes with a

few real CC-BY series; the synthetic cases isolate regimes and the controls guard against leakage, but the real-data coverage is reported on a handful of public series, not a large corpus. Results are shown at horizon one and nominal 80%; the harness supports other horizons and levels. The value we claim is the reusable stateful-streaming harness and the leakage-guarded measurement, not a new calibrator.

VII. CONCLUSION

We built and released `preqts`, a prequential streaming evaluation harness for stateful time-series forecasters with an explicit covariate-arrival policy, embedded in an atlas that pairs ten diagnostic families with a nineteen-method ladder so the diagnosis explains the leaderboard, and used it to show, reproducibly and against negative controls, that online conformal calibration reduces streaming interval coverage error seven-

to-tenfold and collapses its across-case spread while leaving the point forecast unchanged. Evaluating forecasters the way they are deployed, as a stream with state, is what makes the calibration behavior visible; the harness and the released atlas make it reproducible.

DATA AND CODE AVAILABILITY

The harness is the open-source package `preqts` (<https://pypi.org/project/preqts/>). The atlas code, the per-case artifacts (trace, analysis, and streaming trajectories) and the workbench are at https://github.com/fsantibanezleal/CAOS_RES_ChronoScope under the MIT license; an interactive workbench is available at <https://chronoscope.fasl-work.com>. Real datasets (UCI Electricity, Beijing PM2.5, M4 excerpts) are distributed under CC-BY-4.0 with attribution; streaming artifacts carry aggregate trajectories only. The figures are regenerated deterministically from these artifacts by `manuscripts/prequential-streaming-conformal/figures/make_figs.py`.

APPENDIX A THE ATLAS ROSTER

The ten diagnostic families and their principal methods are listed in Section II; each is a verified implementation with its own theory page (equations, DOIs, and a theme-aware figure) in the repository `docs/analysis/`. The nineteen-method forecast ladder spans classical (seasonal-naive, SES, Holt, Holt–Winters, Theta), statistical (AutoARIMA, AutoETS, AutoTheta), gradient-boosted (LightGBM), deep, and stateful foundation models (Chronos, TimesFM). The streaming roster reported here is the naive floor (seasonal-naive), a strong classical live method (Theta) with raw intervals, and that same Theta wrapped in ACI and in Conformal-PID; all four are evaluated by the prequential harness of Section III at nominal coverage 80%, horizon one, and a per-case warmup of twice the seasonal period floored at 20 steps (20, 24 or 48 across the cases), on all fifteen cases.

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